
NYCU IMF - 114 Financial Econometrics

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Assignment: 7, Ch11 q24, 28

Question 24

Reconsider Example 11.2 on the supply and demand for fish at the Fulton Fish Market. The data are in the file *fultonfish*.

- (a) Add the variable *MIXED*, which indicates poor but not *STORMY* weather conditions, to the supply equation in equation (11.14). Estimate the new reduced-form equation for $\ln(PRICE)$, adding the variable *MIXED* to equation (11.16). Is it statistically significant at the 5% level? Test the joint significance of *STORMY* and *MIXED*. Is the resulting F-value greater than 10?

The new reduced-form equation for $\ln(PRICE)$ will then be:

$$\begin{aligned}\ln(PRICE_t) = & \pi_{12} + \pi_{22}MON_t + \pi_{32}TUE_t + \pi_{42}WED_t + \pi_{52}THU_t \\ & + \pi_{62}STORMY_t + \pi_{72}MIXED_t + v_{t2}.\end{aligned}$$

1. Is the variable *MIXED* statistically significant at the 5% level?

Yes, from the regression results, the p -value for *MIXED* is 0.00321. Because $0.00321 < 0.05$, we reject the null hypothesis that its coefficient is zero. The variable *MIXED* is highly statistically significant at the 5% level. The coefficient estimate as $\hat{\pi}_{72} = 0.237$ with a standard error of 0.079 and $t = 3.017$.

2. Is the resulting F-value greater than 10?

Yes. We evaluate the joint null hypothesis $H_0 : \pi_{62} = 0, \pi_{72} = 0$. The resulting F -statistic is 16.14 (with a p -value of 7.857×10^{-7}). Because $16.14 > 10$, we can conclude that *STORMY* and *MIXED* are jointly significant. More importantly for 2SLS, this satisfies the rule of thumb for instrumental variables: an F -statistic strictly greater than 10 on the excluded instruments in the first-stage regression indicates that they are strong instruments, meaning we shouldn't have to worry about weak instrument bias when estimating the demand equation.

- (b) Estimate the demand equation using *STORMY* and *MIXED* as IVs. Compare the coefficient estimates to those in Table 11.5.

We then estimate the structural demand equation, instrumenting the endogenous $\ln(PRICE)$ with the supply shifters *STORMY* and *MIXED*.

$$\ln(QUAN_t) = \alpha_1 + \alpha_2 \ln(PRICE_t) + \alpha_3 MON_t + \alpha_4 TUE_t + \alpha_5 WED_t + \alpha_6 THU_t + e_{dt}$$

Comparison of 2SLS Estimates

After estimating the overidentified 2SLS model, we compare these new coefficient estimates to the original ones in Table 11.5.

Table 1: Updated 2SLS Estimates for Fish Demand (Instruments: *STORMY*, *MIXED*)

Variable	Coefficient	Std. Error	<i>t</i> -Statistic	Prob.
<i>C</i>	8.5402	0.1566	54.5330	0.0000
$\ln(PRICE)$	-0.9301	0.3534	-2.6320	0.0098
<i>MON</i>	-0.0119	0.2084	-0.0570	0.9546
<i>TUE</i>	-0.5258	0.2023	-2.5990	0.0107
<i>WED</i>	-0.5626	0.2070	-2.7190	0.0077
<i>THU</i>	0.0999	0.2029	0.4920	0.6235

1. The Price Elasticity of Demand ($\ln(Price)$)

- Original Estimate (Table 11.5): -1.1194 (Standard Error: 0.4286)
- New Estimate: -0.9301 (Standard Error: 0.3534)
- Conclusion: The estimated price elasticity of demand changed from approximately -1.12 to -0.93, making the demand curve appear slightly less elastic. More importantly, notice that the standard error decreased from 0.4286 to 0.3534. This is exactly what we hope to see when adding another valid instrument: it improves the efficiency (precision) of the IV estimator.

2. The Day of the Week Indicators (*MON*, *TUE*, *WED*, *THU*)

- Comparison: If we look at the coefficients for the days of the week, they remained very stable. For example, *TUE* changed minimally from -0.5308 to -0.5258, and *WED* changed from -0.5664 to -0.5626.
- Conclusion: The fixed demand shifts associated with different days of the week are robust and largely unaffected by the inclusion of the second instrument.

(c) Test the validity of the surplus instrument using the Sargan test, discussed in Section 10.3.4.

To test the validity of the surplus instrument, we examine the Sargan test statistic from the 2SLS diagnostics. The null hypothesis for this test is that all instruments are valid (i.e., uncorrelated with the structural error term).

Based on the estimation output:

- Sargan $\chi^2(1)$ statistic = 0.772
- *p*-value = 0.380

Since the p -value (0.380) is greater than 0.05, we fail to reject the null hypothesis at the 5% significance level. We conclude that the instruments (*STORMY* and *MIXED*) are valid and properly excluded from the structural demand equation.

- (d) In the reduced-form equation in part (a), test the joint significance of the indicator variables *MON*, *TUE*, *WED*, and *THU* at the 5% level. What do you conclude? Are we now able to estimate the supply equation by 2SLS with confidence in our procedure?

To determine if we can confidently estimate the structural supply equation using 2SLS, we must test whether the demand shifters (*MON*, *TUE*, *WED*, *THU*) serve as strong instruments for the endogenous variable, $\ln(\textit{PRICE})$. We test their joint significance in the reduced-form price equation.

Based on the linear hypothesis test output:

- F -statistic = 0.6279
- p -value = 0.6437

Because the p -value (0.6437) is greater than 0.05, we fail to reject the null hypothesis that the coefficients on the day-of-the-week variables are jointly zero. They are not statistically significant at the 5% level.

Conclusion: We **cannot** estimate the supply equation by 2SLS with confidence. For instruments to be considered strong, the F -statistic on the excluded instruments in the first stage should be strictly greater than 10. With an F -statistic of only 0.6279, these instruments are extremely weak. This implies that daily demand shifts do not create enough exogenous variation in price to reliably trace out and identify the supply curve.

Question 28

Supply and demand curves as traditionally drawn in economics principles classes have price (P) on the vertical axis and quantity (Q) on the horizontal axis.

- (a) Rewrite the truffle demand and supply equations in (11.11) and (11.12) with price P on the left-hand side. What are the anticipated signs of the parameters in this rewritten system of equations?

To put price (P) on the left-hand side (vertical axis), we need to algebraically solve Equations (11.11) and (11.12) for P_i .

1. The Inverse Demand Equation

Starting with $Q_i = \alpha_1 + \alpha_2 P_i + \alpha_3 P S_i + \alpha_4 D I_i + e_{di}$, we isolate P_i :

$$P_i = -\frac{\alpha_1}{\alpha_2} + \frac{1}{\alpha_2} Q_i - \frac{\alpha_3}{\alpha_2} P S_i - \frac{\alpha_4}{\alpha_2} D I_i - \frac{1}{\alpha_2} e_{di}$$

To make this cleaner to estimate and discuss, we can define new parameters (let's use γ):

$$P_i = \gamma_1 + \gamma_2 Q_i + \gamma_3 PS_i + \gamma_4 DI_i + v_{di}$$

Anticipated Signs for Inverse Demand:

- γ_2 (Coefficient on Q_i): We expect $\gamma_2 < 0$. The law of demand states that $\alpha_2 < 0$ (higher price means lower quantity). Therefore, $\frac{1}{\alpha_2}$ must also be negative. This gives us the downward-sloping demand curve.
- γ_3 (Coefficient on PS_i): We expect $\gamma_3 > 0$. PS is the price of a substitute. An increase in PS increases demand ($\alpha_3 > 0$). Since α_2 is negative, the ratio $-\frac{\alpha_3}{\alpha_2}$ is positive. A higher substitute price shifts the inverse demand curve up.
- γ_4 (Coefficient on DI_i): We expect $\gamma_4 > 0$. Assuming truffles are a normal good, higher disposable income increases demand ($\alpha_4 > 0$). Again, $-\frac{\alpha_4}{\alpha_2}$ becomes positive.

2. The Inverse Supply Equation

Starting with $Q_i = \beta_1 + \beta_2 P_i + \beta_3 PF_i + e_{si}$, we isolate P_i :

$$P_i = -\frac{\beta_1}{\beta_2} + \frac{1}{\beta_2} Q_i - \frac{\beta_3}{\beta_2} PF_i - \frac{1}{\beta_2} e_{si}$$

Let's define new parameters for the supply equation (let's use θ):

$$P_i = \theta_1 + \theta_2 Q_i + \theta_3 PF_i + v_{si}$$

Anticipated Signs for Inverse Supply:

- θ_2 (Coefficient on Q_i): We expect $\theta_2 > 0$. The law of supply dictates $\beta_2 > 0$. Therefore, $\frac{1}{\beta_2}$ is positive, giving us the standard upward-sloping supply curve.
 - θ_3 (Coefficient on PF_i): We expect $\theta_3 > 0$. PF is the price of a production factor (input cost). Higher costs decrease the quantity supplied at any given price ($\beta_3 < 0$). Because $\beta_2 > 0$ and $\beta_3 < 0$, the ratio $-\frac{\beta_3}{\beta_2}$ becomes positive. An increase in factor prices shifts the inverse supply curve up (meaning producers require a higher price to supply the same quantity).
- (b) Using the data in the file *truffles*, estimate the supply and demand equations that you have formulated in (a) using two-stage least squares. Are the signs correct? Are the estimated coefficients significantly different from zero?

For both equations, we are checking if the estimated signs align with standard economic theory and if the coefficients are statistically significant (usually determined by a p -value < 0.05).

1. The Inverse Demand Equation

The estimated equation is:

$$P = -11.428 - 2.671Q + 3.461PS + 13.390DI$$

- Quantity (Q): The coefficient is -2.671 . The sign is correct (negative), reflecting the downward-sloping law of demand. It is statistically significant ($p = 0.0315$).
- Price of Substitute (PS): The coefficient is 3.461 . The sign is correct (positive), meaning an increase in the price of the substitute shifts the demand curve up/right. It is statistically significant ($p = 0.0046$).
- Disposable Income (DI): The coefficient is 13.390 . The sign is correct (positive), assuming truffles are a normal good. It is highly significant ($p < 0.001$).

2. The Inverse Supply Equation

The estimated equation is:

$$P = -58.798 + 2.9367Q + 2.9585PF$$

- Quantity (Q): The coefficient is 2.9367 . The sign is correct (positive), reflecting the upward-sloping law of supply. It is highly significant ($p < 0.001$).
- Price of Factor (PF): The coefficient is 2.9585 . The sign is correct (positive), meaning an increase in production costs shifts the inverse supply curve up (producers require a higher price to supply the same quantity). It is highly significant ($p < 0.001$).

(c) Estimate the price elasticity of demand “at the means” using the results from (b).

The price elasticity of demand is defined as:

$$E_d = \frac{dQ}{dP} \times \frac{\bar{P}}{\bar{Q}}$$

Because we estimated the inverse demand equation in part (b), the estimated coefficient on quantity (Q) represents $\frac{dP}{dQ} = -2.671$. Therefore, the marginal effect of price on quantity is the inverse of this coefficient:

$$\frac{dQ}{dP} = \frac{1}{-2.671} \approx -0.3744$$

Using the sample means provided ($\bar{P} = 62.724$ and $\bar{Q} = 18.45833$), we calculate the elasticity:

$$\begin{aligned} E_d &= \left(\frac{1}{-2.671} \right) \times \left(\frac{62.724}{18.45833} \right) \\ E_d &\approx -0.3744 \times 3.3981 \\ E_d &\approx -1.2725 \end{aligned}$$

Conclusion: The estimated price elasticity of demand at the sample means is approximately -1.2725 . Because the absolute value is greater than 1 ($|-1.2725| > 1$), the demand for truffles is considered elastic at the average price and quantity levels.

- (d) Accurately sketch the supply and demand equations, with P on the vertical axis and Q on the horizontal axis, using the estimates from part (b). For these sketches set the values of the exogenous variables DI , PS , and PF to be $DI^* = 3.5$, $PF^* = 23$, and $PS^* = 22$.

To sketch the equations in a standard P - Q space, we substitute the given exogenous values ($DI^* = 3.5$, $PS^* = 22$, and $PF^* = 23$) into our 2SLS estimates from part (b).

Inverse Demand Curve:

$$\begin{aligned} P &= -11.428 - 2.671Q + 3.461(PS) + 13.390(DI) \\ P &= -11.428 - 2.671Q + 3.461(22) + 13.390(3.5) \\ P &= 111.579 - 2.671Q \end{aligned}$$

This yields a downward-sloping demand curve with a price intercept of 111.579.

Inverse Supply Curve:

$$\begin{aligned} P &= -58.798 + 2.9367Q + 2.9585(PF) \\ P &= -58.798 + 2.9367Q + 2.9585(23) \\ P &= 9.2475 + 2.9367Q \end{aligned}$$

This yields an upward-sloping supply curve with a price intercept of 9.2475.

Setting the two equations equal to each other allows us to find the equilibrium coordinates for our sketch:

$$\begin{aligned} 111.579 - 2.671Q &= 9.2475 + 2.9367Q \\ 102.3315 &= 5.6077Q \\ Q^* &\approx 18.248 \end{aligned}$$

Substituting Q^* back into the demand equation yields the equilibrium price:

$$P^* = 111.579 - 2.671(18.248) \approx 62.839$$

- (e) What are the equilibrium values of P and Q obtained in part (d)? Calculate the predicted equilibrium values of P and Q using the estimated reduced-form equations from Table 11.2, using the same values of the exogenous variables. How well do they agree?

1. Values obtained from the 2SLS structural equations in part (d):

- $Q^* \approx 18.248$
- $P^* \approx 62.839$

2. Values obtained from the reduced-form equations (Table 11.2):

Using the same exogenous values ($DI^* = 3.5$, $PS^* = 22$, and $PF^* = 23$), we substitute them into the reduced-form estimates.

For Quantity (Q):

$$Q = 7.8951 + 0.6564(PS) + 2.1672(DI) - 0.5070(PF)$$

$$Q = 7.8951 + 0.6564(22) + 2.1672(3.5) - 0.5070(23)$$

$$Q \approx 18.260$$

For Price (P):

$$P = -32.5124 + 1.7081(PS) + 7.6025(DI) + 1.3539(PF)$$

$$P = -32.5124 + 1.7081(22) + 7.6025(3.5) + 1.3539(23)$$

$$P \approx 62.814$$

Conclusion: The predicted equilibrium values from the 2SLS structural equations ($Q^* \approx 18.248$, $P^* \approx 62.839$) and the OLS reduced-form equations ($Q^* \approx 18.260$, $P^* \approx 62.814$) agree almost perfectly. The extremely minor discrepancies (at the hundredths decimal place) are entirely attributable to rounding errors in the textbook's reported tables and our intermediate calculations. Theoretically, both methods should trace the exact same equilibrium intersection.

- (f) Estimate the supply and demand equations that you have formulated in (a) using OLS. Are the signs correct? Are the estimated coefficients significantly different from zero? Compare the results to those in part (b).

Estimating with OLS and Comparison to 2SLS

When we estimate the structural equations using ordinary least squares (OLS), we ignore the fact that price (P) and quantity (Q) are determined simultaneously in equilibrium. This simultaneity means that Q is an endogenous variable; it is correlated with the error terms in both the supply and demand equations, violating the core OLS assumption ($E[e|X] = 0$).

Comparing the Results:

- **Demand Equation Breakdown:** The OLS estimation of the demand curve is severely biased. The coefficient on quantity (Q) flipped from a statistically significant -2.671 in 2SLS to an insignificant $+0.1512$ in OLS. OLS completely failed to identify the downward-sloping demand curve, yielding a theoretically nonsensical positive slope.
- **Supply Equation Breakdown:** The OLS estimation of the supply curve fared slightly better in terms of signs. Both 2SLS and OLS yielded positive, significant coefficients for Q (2.9367 vs. 2.6613). However, the OLS estimate is still biased downward due to endogeneity.

Conclusion: The OLS estimates suffer from severe simultaneous equations bias (endogeneity bias). Because equilibrium points are generated by shifts in both supply and demand, OLS cannot isolate the individual curves. The 2SLS procedure used in part (b) successfully corrects for this bias by using instrumental variables (PF for demand, and PS, DI for supply) to isolate the exogenous variation in Q , allowing us to consistently estimate the true structural parameters.